

Toward Size-Robust Zero-Shot Search for Binary Linear Systems:

v7 — Expert Imitation, Test-Time Iteration Scaling, and What the Ablations Attribute

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Abstract

Motivated by a literature review of zero-shot generalization in reinforcement learning and neural combinatorial optimization, the v7 system replaced self-play with imitation of a propagation-impact expert, trained on a discrete-uniform instance-size distribution ($m \sim U\{6, \dots, 20\}$, $n = 2.5m$) with state-dependent size-invariant features, and decoupled the recurrent GNN’s message-passing iteration count so it can be scaled at inference time. Evaluation followed a strict zero-shot protocol: training never saw $m > 20$, and 40×100 was evaluated as pure size extrapolation. Three attributions emerge from matched ablations, reported with timing throughout. First, imitation of a fail-first expert perfected the smallest size — 100% feasibility and 100% solution accuracy at 10×25 (1.52 s), removing all seven residual budget false positives of v6 by shortening infeasibility proofs (DFS infeasibility proofs rose from 48 to 55) — the first non-exact method in the project’s benchmark to saturate this size (the best inexact baseline, BKZ, reaches 72.1% solution accuracy there; v6 report, Table 2). Second,

the same imitation *degraded* solution search at 20×50 (5.50% vs. 8.50% for the pre-trained-only checkpoint at identical budgets and iterations), and the iteration count was ruled out as a cause; combined with v6’s self-play result, two independent forms of open-loop policy training have now been shown to harm the pre-trained policy’s search utility at the hard boundary, while helping refutation — consistent with the classical fail-first/succeed-first asymmetry in constraint programming. Third, test-time iteration scaling ($8 \rightarrow 32$ rounds at 40×100) changed nothing: zero-shot extrapolation remains pump-bound at 2.25% solution accuracy, indicating that no adjustment of the current network’s inference regime bridges the gap, and reinforcing the previously identified roadmap (conflict-driven search and lattice-basis enumeration) as the binding path forward.

1 Design, from the Literature

Five lessons were extracted from the survey (`ref/zero-shot-size-generalization-rl.md`) and mapped to v7: imitation rather than self-play as the size-transfer signal (Gasse et al., 2019); test-time iteration scaling of a weight-shared recurrent GNN (Selsam et al., 2019); explicit size-normalized nonlinear features, since ReLU networks extrapolate linearly (Xu et al., 2021) and size-dependent local statistics break GNN size generalization (Yehudai et al., 2021); feature construction aligned with the propagation algorithm (Xu et al., 2020); and a randomized training-size distribution in the spirit of procedural-generation benchmarks (Cobbe et al., 2020). The expert is a propagation-impact oracle: for each unassigned variable both values are probed, contradictions rank highest (fail-first), and propagation-fixed counts break ties; half of the imitation rollouts follow the planted solution for assignment targets, half follow the expert greedily. Training used masked-solution pre-training (60 epochs, mask accuracy 79.2%) followed by 20 imitation epochs, entirely on fresh instances with $m \in [6, 20]$.

Table 1: v7 results with matched v6 references. Feas = feasibility accuracy (%), Sol = solution accuracy (%), t = average seconds per instance including preprocessing. The 40×100 column is zero-shot for v7 (size never trained); v6 trained on all three sizes.

Model	10×25			20×50			40×100 (v7: zero-shot)		
	Feas	Sol	t	Feas	Sol	t	Feas	Sol	t
v6 (pretrain, best)	99.3	100.0	1.50	82.5	9.6	7.12	80.2	2.25	25.0
v7 (pretrain + imitation)	100.0	100.0	1.52	82.5	5.5	10.8	80.2	2.25	31.5

2 Zero-Shot Protocol and Results

Training never saw an instance with $m > 20$. Test sets are the hard-family sets of the v6 cycle: full 1,000-instance sets at 10×25 and 20×50 (both inside the training size range) and the stratified 500-instance subset at 40×100 (pure extrapolation). Search budgets match v6 (50 simulations, DFS budget $20k \cdot n/25$ nodes); message-passing iterations default to $\lceil 8n/25 \rceil$.

2.1 Attribution ablations at 20×50

The 5.5% headline conflates three design changes; matched ablations separate them (all at 50 simulations on the same 1,000 instances). Running the imitation checkpoint at the *trained* iteration count (8 instead of 16) leaves solution accuracy unchanged at 5.50% (8.10 s), ruling out iteration-count extrapolation as the cause. Running the v7 *pre-trained-only* checkpoint at the same settings yields 8.50% (6.27 s), close to the v6 pre-trained reference (9.62%); the residual gap is plausibly due to the size-distribution pre-training spreading exposure across fifteen sizes, though this attribution is a single-seed point estimate. All comparisons in this section are single runs; the main regression (44 vs. 68 solved of 800) is individually significant ($z \approx 2.4$) but multi-seed confirmation is future work. The regression is therefore caused by the imitation phase itself.

2.2 The fail-first asymmetry

The same imitation that harms solution search at 20×50 *perfected* 10×25 : v6’s seven residual false positives — infeasible instances whose refutation exhausted the DFS budget — are all closed by v7 (proofs $48 \rightarrow 55$, false positives $7 \rightarrow 0$), because a fail-first ordering shortens proof trees. This is the classical constraint-programming asymmetry: fail-first branching is the right heuristic for refutation, while solution-finding on satisfiable instances favors succeed-first guidance. The expert’s contradiction-dominated scoring taught the former at the expense of the latter. Together with v6’s self-play result, two independent open-loop policy-training signals have now degraded the pre-trained policy’s search utility at the hard boundary; the masked-prediction prior remains the strongest solution-search policy tested.

2.3 Test-time iteration scaling: a null result

At 40×100 , scaled iterations (32) and trained iterations (8) produce identical zero-shot outcomes (80.2% / 2.25%), with an unexplained $\sim 11\%$ time difference between the variants (31.5 s vs. 35.3 s) that per-stage node logging, absent in the current harness, would be needed to attribute. All nine solved instances come from the kernel pump in both configurations. The NeuroSAT-style mechanism (Selsam et al., 2019) does not transfer to this architecture and problem: whatever additional message passing computes, the policy it feeds cannot exploit at four times the trained size. Zero-shot extrapolation remains bounded by the pump, whose flat cost curve and size-independent contribution are again the pipeline’s only scale-robust element.

3 Consequences

Three cycles of evidence now triangulate the same conclusion. Learned open-loop policies — whether self-play-trained (v6) or expert-imitated (v7) — do not carry solution search across the hardness boundary, and inference-regime adjustments (budget in v6, iterations

in v7) do not either. What does work is structural: propagation, the null-space pump, and complete search whose ordering matters differently for proofs (fail-first, now learned and paying off at 10×25) than for solutions. The reinforcement roadmap of the v6 report is therefore unchanged in content but sharpened in emphasis: (1) conflict-driven search (no-good learning) attacks the proof side where learned orderings already demonstrably help; (2) lattice-basis enumeration attacks the solution side where every learned and budget-based lever has now been exhausted; (3) if policy training is attempted again, it should target *proof-tree size* on infeasible instances — the one objective where imitation has produced a measurable, transferable gain — rather than solution construction. A per-size deployment recommendation follows from Table 1 at equal honesty: the imitation checkpoint where proofs dominate (small sizes), the pre-trained checkpoint where search dominates (larger sizes); a principled selector between them is future work, and the split itself is evidence that ordering policy is not one-size-fits-all.

References

- Cobbe, K., Hesse, C., Hilton, J., & Schulman, J. (2020). Leveraging procedural generation to benchmark reinforcement learning. *ICML*.
- Gasse, M., Chételat, D., Ferroni, N., Charlin, L., & Lodi, A. (2019). Exact combinatorial optimization with graph convolutional neural networks. *NeurIPS* 32.
- Selsam, D., Lamm, M., Bünz, B., Liang, P., de Moura, L., & Dill, D. L. (2019). Learning a SAT solver from single-bit supervision. *ICLR*.
- Xu, K., Li, J., Zhang, M., Du, S. S., Kawarabayashi, K., & Jegelka, S. (2020). What can neural networks reason about? *ICLR*.
- Xu, K., Zhang, M., Li, J., Du, S. S., Kawarabayashi, K., & Jegelka, S. (2021). How neural networks extrapolate: From feedforward to graph neural networks. *ICLR*.

Yehudai, G., Fetaya, E., Meirom, E., Chechik, G., & Maron, H. (2021). From local structures to size generalization in graph neural networks. *ICML*.